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Modern Communication Systems F25

Lab 5

11/10/2025

1) Receive Diversity with MRC

a) Vector model (1 line)

$$\mathbf{y}[k] = [y_1[k] \ \ y_2[k]]$$

$$\mathbf{h} = [h_1 \ \ h_2]$$

$$\mathbf{n}[k] = [n_1[k] \ \ n_2[k]]$$

b) MRC decision statistic

non-zero scalar multiples of $\mathbf{h}$, $\mathbf{w} = \mathbf{h}$

$$\hat{x}[k] = \mathbf{h}^H \mathbf{y}[k] = h_1^* y_1[k] + h_2^* y_2[k]$$

c) Post-combiner SNR

with $\mathbf{w} = \mathbf{h}$,

$$\text{signal term at combiner output: } (\mathbf{h}^H \mathbf{h}) x[k] = (|h_1|^2 + |h_2|^2) x[k]$$

$$\text{signal power: } (|h_1|^2 + |h_2|^2)^2 P$$

$$\text{noise variance at output: } N_0 \|\mathbf{h}\|^2 = N_0 (|h_1|^2 + |h_2|^2)$$

$$\gamma_{\text{MRC}} = \text{signal power} / \text{noise power} = (|h_1|^2 + |h_2|^2)^2 P / N_0 (|h_1|^2 + |h_2|^2) = (|h_1|^2 + |h_2|^2) P / N_0$$

d) Numerical example

i)

$$|h_1|^2 = (1)^2 + (1)^2 = 2$$

$$|h_2|^2 = (.5)^2 + (-1)^2 = 1.25$$

$$\gamma_{\text{MRC}} = ((2) + (1.25)) / (.5) = 6.5$$

ii)

$$\gamma_1 = ((2)(1)) / (.5) = 4$$

$$\gamma_2 = ((1.25)(1)) / (.5) = 2.5$$

e) Selection combining (compare)

$$\gamma_{\text{SC}} = \max(\gamma_1, \gamma_2) = \max(4, 2.5) = 4$$

$$\gamma_{\text{MRC}} / \gamma_{\text{SC}} = 6.5 / 4 = 1.625$$

or about 2.11dB

so MRC gives about 2.11dB improvement over selection combining

f) Quick BER check (BPSK)

i)

$$\gamma = 6.5$$

$$P_{\text{bMRC}} = Q(\sqrt{13})$$

ii)

$$\gamma = 4$$

$$P_{\text{bMRC}} = Q(\sqrt{8}) = Q(2\sqrt{2})$$

2) Transmit Diversity MRC Beamforming vs. Alamouti

a) MRC beamforming with CSIT (derivation)

i)

$$y = \mathbf{h}^T \mathbf{w} s + n$$

$$\text{maximize } |\mathbf{h}^T \mathbf{w}|^2 \text{ subject to } \|\mathbf{w}\| = 1$$

$$\mathbf{w} = \mathbf{h}^* / \|\mathbf{h}\|$$

ii)

$$|\mathbf{h}^T \mathbf{w}|^2 = |\mathbf{h}^T \mathbf{h}^* / \|\mathbf{h}\| |^2 = |\mathbf{h}^T \mathbf{h}^*|^2 / \|\mathbf{h}\|^2 = \|\mathbf{h}\|^2$$

$$\gamma_{\text{BF}} = \|\mathbf{h}\|^2 P / N_0 = (|h_1|^2 + |h_2|^2) P / N_0$$

b) naïve repeat without CSIT

i)

$$y = h_1 (\sqrt{1/2}) s + h_2 (\sqrt{1/2}) s + n = \sqrt{1/2} (h_1 + h_2) s + n$$

$$\gamma_{\text{rep}} = |h_1 + h_2|^2 P / 2N_0$$

ii)

when h_1 and h_2 have nearly opposite phases so $h_1 + h_2$ is near 0

c) Alamouti STBC (no CSIT)

i) a

$$y_t = h_1 x_1 + h_2 x_2 + n_t$$

$$y_{t+1} = h_1 (-x_2^*) + h_2 x_1^* + n_{t+1} = -h_1 x_2^* + h_2 x_1^* + n_{t+1}$$

ii) a

$$x_1 = h_1^* y_t + h_2 y_{t+1}^*$$

$$x_1 = h_1^* (h_1 x_1 + h_2 x_2 + n_t) + h_2 (-h_1 x_2^* + h_2 x_1^* + n_{t+1})$$

$$= (|h_1|^2 + |h_2|^2) x_1 + (h_1^* h_2 n_t + h_2 h_1^* n_{t+1}^*)$$

x_2 cancels so each symbol sees effective channel gain $|h_1|^2 + |h_2|^2$

iii) a

$$E[|x_1|^2] = E[|x_2|^2] = P/2$$

$$\gamma_{\text{Alam}} = (|h_1|^2 + |h_2|^2) P / 2N_0$$

d) Numerical comparison (use the same channel for all parts)

i) Compute $|h_1|^2$, $|h_2|^2$, and $\|\mathbf{h}\|^2$

$$|h_1|^2 = 0.6^2 + 0.8^2 = 1$$

$$|h_2|^2 = (-0.3)^2 + 0.4^2 = 0.25$$

$$\|\mathbf{h}\|^2 = |h_1|^2 + |h_2|^2 = 1.25$$

ii) CSIT beamforming

$$\gamma_{BF} = \|h\|^2 P / N_0 = 1.25 * 1 / .2 = 6.25$$

7.96dB

iii) naïve repeat

$$h_1 + h_2 = .3 + 1.2j$$

$$|h_1 + h_2|^2 = 1.53$$

$$\gamma_{rep} = |h_1 + h_2|^2 P / 2N_0 = 1.53 * 1 / (2 * .02) = 3.825$$

5.82dB

loss of 2.14dB

iv) Alamouti

$$\gamma_{Alam} = \|h\|^2 P / 2N_0 = 1.25 / (2 * .2) = 3.125$$

4.95dB

loss of 3.01

e) Explanation

Alamouti guarantees each symbol gets a combined power $|h_1|^2 + |h_2|^2$ regardless of relative phases of h_1 and h_2 . The naïve approach can suffer from destructive interference when phases are opposite.

3) Simple 2×2 MIMO with Zero-Forcing

a) ZF equalizer and effective noise (one-step derivation).

$$\hat{\mathbf{x}} = \mathbf{W}\mathbf{y} = \mathbf{H}^{-1}(\mathbf{H}\mathbf{x} + \mathbf{n}) = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

$$\hat{x}_1 = [\mathbf{H}^{-1}]_{11} y_1 + [\mathbf{H}^{-1}]_{12} n_2$$

$$\hat{x}_2 = [\mathbf{H}^{-1}]_{21} y_1 + [\mathbf{H}^{-1}]_{22} n_2$$

effective noise on stream i is $[\mathbf{H}^{-1}]_{i2} n_2$

since $n_2 \sim \mathcal{CN}(0, N_0)$, the noise variance on stream i is $N_0 \|[\mathbf{H}^{-1}]_{i2}\|^2$

thus the SNR for stream i is $\gamma_i = P / (N_0 \|[\mathbf{H}^{-1}]_{i2}\|^2)$

b) Numeric evaluation.

i) a

$$\det(\mathbf{H}) = 1 \cdot 1.2 - .5 \cdot (-.3) = 1.2 + .15 = 1.35$$

$$\mathbf{H}^{-1} = 1/1.35 [1.2 \ \ .5 \ \ .3 \ \ 1]$$

$$= [.888888 \ \ -.370370 \ \ .222222 \ \ .740740]$$

ii) a

$$\mathbf{w}_1 = [.8888 \ \ -.3703]$$

$$\|\mathbf{w}_1\|^2 = .8888^2 + (-.3703)^2 = .927297$$

$$\|\mathbf{w}_2\|^2 = .2222^2 + .7407^2 = .59807$$

iii) a

$$\gamma_i = P / (N_0 \|\mathbf{w}_i\|^2)$$

$$\gamma_1 = 1 / (.2 \cdot .927) = 5.3920$$

$$\gamma_2 = 1 / (.2 \cdot .598) = 8.3602$$

iv) a

stream 1 is more noise enhanced with lower SNR

-1.91dB

c) Explanation

ZF suffers when $\mathbf{H}$ is ill conditioned and the rows/cols are nearly linearly dependent ($\det(\mathbf{H})$ is small)

4) MIMO Throughput for an 802.11n-like Link

a) Compute SNRs (dB) and choose MCS

$$\gamma_{\text{SISO}} = |h_1|^2 P/N_0 = .4 * 10 / .1 = 40 = 16.02 \text{ dB}$$

best is 14dB at 16-QAM $R = 1/2$

$$\gamma_{\text{MRC}} = (|h_1|^2 + |h_2|^2) P/N_0 = (.4 + .8) * 10 / .1 = 120 = 20.79 \text{ dB}$$

best is 18dB at 16-QAM $R = 3/4$

b) Throughput

$$\text{SISO: bits/symbol} = 52 * 4 * 1/2 = 104$$

$$\text{throughput} = 104 * 250000 = 26 \text{ Mbps}$$

$$\text{MRC: bits/symbol} = 52 * 4 * 3/4 = 156$$

$$\text{throughput} = 156 * 250000 = 39 \text{ Mbps}$$

1.5, a 50% improvement

c) Explanation

MRC coherently sums the powers from both antennas, increasing instantaneous SNR, allowing a higher rate MCS and more throughput

d) Explanation

If MCS is smaller, the throughput will simply decrease with fewer bits per symbol. The BER may improve, more reliability, but diminishing returns.

If MCS is larger, there will be higher BER, more packets needing to be resent, and lower net throughput despite the increase in bits per symbol.

5) Programming: Rank, Invertibility, and SVD (2×2)

a) Singular values, rank, invertibility

i) H1

$$H^TH = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

eigenvalues 4, 0

$$(\sigma_1, \sigma_2) = (2, 0)$$

rank=1

det=0

invertible? no

ii) H2

$$H^TH = \begin{bmatrix} 1 & -0.2 \\ 0.6 & 1.1 \end{bmatrix} \begin{bmatrix} 1 & 0.6 \\ -0.2 & 1.1 \end{bmatrix} = \begin{bmatrix} 1.04 & 0.38 \\ 0.38 & 1.57 \end{bmatrix}$$

eigenvalues 1.768, 0.842

$$(\sigma_1, \sigma_2) = (1.330, 0.917)$$

rank=2

det=1.4884

invertible? yes

iii) H3

$$H^TH = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

eigenvalues 1.105, 0.904

$$(\sigma_1, \sigma_2) = (1.0512, 0.951)$$

rank=

det=1

invertible? yes

b) Condition number and ZF sensitivity

i) H2

ii) H3

c) Explanation